

UNIVERSITÀ DEGLI STUDI DI MILANO-BICOCCA

STAGE

A survey on Information Diffusion Models

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1 Introduction

The spread of information on social networks is an extraordinarily important contemporary phenomenon. The process by which information propagates from an initial node through the connections of a network follows dynamics that can be formalized and analyzed using mathematical models. The ability to understand and predict how information spreads is crucial for numerous practical applications: ranging from preventing the spread of misinformation to modeling the adoption of technological innovations, to applications in marketing to epidemiology. Information diffusion models can be divided into two broad categories: explanatory models and predictive models. While explanatory models operate retrospectively, seeking to reconstruct the hidden mechanisms that determined an observed diffusion, predictive models aim to anticipate how future processes will proceed.

2 Preliminary Definitions

- **Social network**

A social network can be formally represented as a graph $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_n\}$ is the set of nodes (representing users) and $E \subseteq V \times V$ is the set of directed edges (representing social relationships). An edge $(v_i, v_j) \in E$ indicates that user v_i follows, is friends with, or is otherwise connected to v_j , and may therefore receive information from them. According to Kumar and Sinha [1], network topology plays a crucial role in determining the rate and intensity of diffusion, since the density of connections significantly influences the way information spreads.

- **Activation Sequence**

An activation sequence is an ordered set of nodes $\sigma = (u_1, u_2, \dots, u_k)$ where $u_i \in V$ represents the i -th node that adopts a specific piece of information. Each node u_i is associated with a timestamp t_i that records the time of adoption, with $t_1 < t_2 < \dots < t_k$.

- **Diffusion Cascade**

A diffusion cascade is a directed graph $C = (V_C, E_C)$ where $V_C \subseteq V$ is the set of nodes that have adopted the information and E_C explicitly encodes the influence relationships: an arc $(v_i, v_j) \in E_C$ indicates that node v_i directly influenced node v_j in the adoption process. The cascade is typically represented as a tree rooted in the source node v_s that originally introduced the information.

- **Social Influence**

Social influence is a phenomenon by which actions taken by a user induce their direct contacts to behave in a similar way. Formally, the probability that a node v_j will adopt a piece of information increases if at least one of its neighbors v_i has already adopted it.

- **Herd Behavior**

Herd behavior occurs when a sequence of individuals performs identical actions, sometimes disregarding their own independent information signals. Mathematically, this is expressed when the probability of adoption for node v_i depends primarily on the actions of its predecessors, rather than on endogenous evaluations.

- **Information Cascade**

An information cascade is a phenomenon in which the adoption of information by a set of nodes results exclusively from the observation of others' choices, regardless of the private information signals available to the nodes themselves. This occurs when the flow of public information exceeds the value of private information in the decision-making process. According to Guille et al.[2], these phenomena are intrinsically linked to the processes of information diffusion in online social networks, where users are frequently influenced by the actions of their social contacts.

2.1 Closed World Assumption

All models described below operate under the closed-world assumption information can only propagate through network edges; there is no external influence. Formally:

$$P(\text{node } v_j \text{ adopts} \mid \text{history up to time } t) = P(\text{node } v_j \text{ adopts} \mid \text{neighbors activated up to } t)$$

However, empirical research on Twitter, as pointed out by Guille et al., has shown that this assumption is strongly violated: only 71% of the observed diffusion is due to internal influence, while the remaining 29% comes from external sources. Myers et al., as cited by Guille et al., propose modeling external influence using hazard functions, allowing information to propagate via unobservable external sources.

Another significant limitation, highlighted by Guille et al., concerns the assumption that topology remains static over time. In reality:

$$G(t_1) \neq G(t_2) \text{ for } t_1 \neq t_2$$

Online social networks are highly dynamic, with edges being continuously added and removed.

3 Explanatory models

Epidemiological models are the most established theoretical basis for modeling the spread of information. As Kumar and Sinha highlight, the process of information propagation in a social network can be directly paralleled to the way infectious diseases spread through a population. The conceptual similarity between disease transmission and information diffusion is profound: both processes involve the exposure of a passive (susceptible) agent to an active (infected) agent through network topology.

According to Kumar and Sinha [1], this parallelism is valid because in both epidemiology and information diffusion, the dynamics are governed by:

- A population structured into discrete compartmental states;
- A transition rate between states dependent on system parameters;
- A propagation speed linked to network topology and interactions between individuals

3.1 SIR Model

In the SIR (Susceptible-Infected-Recovered) model, the population is categorized into three states:

- S (Susceptible): individuals not yet exposed to the information;
- I (Infected): individuals currently infected and capable of transmitting the disease;
- R (Recovered): individuals who have recovered and are no longer affected by the information.

State transitions follow ordinary differential equations. Let $S(t)$, $I(t)$, $R(t)$ be the fractions of the population in each state at time t , with $S(t) + I(t) + R(t) = 1$. The dynamics is governed by the following equations:

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

where: $\beta > 0$ is the transmission rate, $\gamma > 0$ is the recovery rate

The first equation shows that the rate of decrease in susceptible individuals is proportional to the product SI , expressing the fact that the number of new infections depends both on the fraction of susceptible individuals and the fraction of infected individuals. The first term of the second equation represents the generation of new infections (identical to the loss term in S), while the second term represents the transition to recovery. The third term quantifies the number of individuals who complete the path to recovery.

The basic reproduction number is defined as:

$$R_0 = \frac{\beta}{\gamma}$$

This parameter represents the average number of susceptible individuals who will be infected by a single infected individual during their infectious period. If $R_0 > 1$, the infection spreads through the population, causing an epidemic; if $R_0 < 1$, the infection dies out naturally. The SIR model exhibits damped oscillator dynamics: the fraction of infected individuals initially grows, reaches a peak, then decreases and stabilizes, with the number of oscillations depending on the parameters.

3.2 SIS Model

In the SIS (Susceptible-Infected-Susceptible) model, as described by Kumar, Sinha, and Guille et al., there are only two states. Unlike SIR, recovered individuals return to the susceptible state (no permanent immunity). This model is particularly appropriate for modeling scenarios in which individuals may be repeatedly exposed to the same agent without developing lasting immunity.

The dynamics is governed by the following equation:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI + \gamma I \\ \frac{dI}{dt} &= \beta SI - \gamma I\end{aligned}$$

with the constraint $S(t) + I(t) = 1$.

In the SIS model, a steady state must be reached, where the flows into and out of each state remain balanced, leading to $\frac{dI}{dt} = 0$:

$$\beta SI = \gamma I$$

Taking into account equilibrium, if $\beta > \gamma$, the steady state has $I > 0$ (endemic equilibrium), indicating that infection persists in the population at a steady level; if $\beta \leq \gamma$, then $I = 0$ (complete eradication of infection).

A modified version that takes into account network heterogeneity, which is crucial when the degree distribution is not uniform, is:

$$I = 1 - \frac{\gamma}{\beta \langle k \rangle}$$

where $\langle k \rangle$ is the average degree of the network. This formula, highlighted by Kumar and Sinha, shows that the diffusion rate critically depends on the distribution of degrees in the network: more densely connected networks (with high average degrees) allow for faster diffusion rates. This reflects the intuition that highly connected individuals act as super-spreaders, accelerating overall propagation.

3.3 SIRS Model

The SIRS (Susceptible-Infected-Recovered-Susceptible) model, as presented in Kumar and Sinha, represents a compromise between SIR and SIS. It models scenarios in which immunity is not permanent but lasts for a limited period, after which the individual becomes vulnerable again:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI + \sigma R \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I - \sigma R\end{aligned}$$

where $\sigma > 0$ is the waning immunity rate.

The parameter σ governs the speed with which individuals who had achieved immunity return to the susceptible state. When $\sigma \approx 0$, the model approaches SIR behavior (quasi-permanent immunity), while when σ is large, the model approximates SIS (very short immunity). The SIRS model therefore represents a more realistic generalization, capturing intermediate scenarios that are common in practice.

3.4 Bass Model

According to Kumar and Sinha [1], the Bass model is an extension of classical epidemiological models designed to model the adoption of technological innovations and products. Unlike SIR/SIS models, the Bass model introduces two distinct mechanisms of influence:

1. Exogenous influence (innovation): p - the probability that a susceptible individual will adopt the innovation regardless of the influence of others (*mass-media influence*);
2. Endogenous influence (imitation): q - the probability that a susceptible individual will adopt the innovation after observing the adoption of others (*word-of-mouth influence*)

The model describes the number of new adopters at time t by:

$$\frac{dN(t)}{dt} = (p + q \cdot n(t)) \cdot (N_{\text{tot}} - N(t))$$

where: - $N(t)$ is the cumulative number of adopters at time t - $n(t) = N(t)/N_{\text{tot}}$ is the fraction of adopters - N_{tot} is the total size of the potential population

The Bass model yields the well-known S-shaped (sigmoid) diffusion curve: diffusion starts slowly, accelerates in the middle phase when imitation dominates, and slows down again in the final stages. Kumar and Sinha point out that the Bass model is particularly powerful for long-term predictions in the adoption of new products and technologies.

3.5 Reversible Dynamics in Real Models

According to Kumar and Sinha [1], a crucial aspect of real-world processes is that not all individuals who enter the “infected” state remain in that state permanently. Their proposed model incorporates reversible transitions, allowing nodes to temporarily adopt and then abandon information or behaviors before eventually adopting them permanently. This reflects the empirical reality where individuals may initially embrace a new product, technology, or idea, find that it does not meet their needs, revert to their previous state, and subsequently, perhaps after adjustments or improvements, adopt the information again permanently.

4 Graph-Based Predictive Models

4.1 Independent Cascades (IC): Sender-Centric Approach

In the Independent Cascades (IC) model, described by Guille et al., each arc $(v_i, v_j) \in E$ is associated with a diffusion probability $p_{ij} \in [0, 1]$. Compared to epidemiological models that operate at an aggregate level on fractions of the population, IC operates on the network, enabling precise tracing of which nodes influence which others.

The process evolves iteratively on a discrete time axis $t = 0, 1, 2, \dots$, starting from a set of initially activated nodes S_0 .

At each time step t :

1. Let $\mathcal{A}(t)$ be the set of nodes activated up to time t and $\mathcal{A}_{\text{new}}(t)$ the nodes activated at time $t - 1$;
2. For each node $v_j \notin \mathcal{A}(t)$, and for each $v_i \in \mathcal{A}_{\text{new}}(t)$ such that $(v_i, v_j) \in E$: - With probability p_{ij} , node v_i activates v_j (the transition occurs only once for each pair of nodes)

The probability that node v_j is activated in step t is therefore:

$$P(v_j) = 1 - \prod_{v_i \in \mathcal{A}_{\text{new}}(t): (v_i, v_j) \in E} (1 - p_{ij})$$

The process is monotone: once activated, a node remains activated and cannot return to the inactive state. The process ends when no new nodes can be activated, i.e., when no neighbor of any active node remains inactive. This model is sender-centric: the success of activation depends primarily on the willingness of the transmitting node to disseminate the information, not on the receiver. It represents situations where information is actively “pushed” to neighbors.

4.2 Linear Threshold (LT): Receiver-Centric Approach

In the Linear Threshold model, instead of probabilities on edges, each edge (v_i, v_j) is assigned an influence weight $w_{ij} \in [0, 1]$ and each node v_j has an influence threshold $\theta_j \in [0, 1]$, usually $\theta_j = 0.5$. The weights must satisfy normalization: $\sum_{v_i: (v_i, v_j) \in E} w_{ij} \leq 1$ for each node v_j .

At each time step t , for each inactive node $v_j \notin \mathcal{A}(t)$:

1. Calculate the accumulated influence: $\Phi_j(t) = \sum_{v_i \in \mathcal{A}(t): (v_i, v_j) \in E} w_{ij}$;
2. If $\Phi_j(t) \geq \theta_j$, then v_j activates and enters $\mathcal{A}(t+1)$

Formally:

$$v_j \in \mathcal{A}(t+1) \text{ if } \sum_{v_i \in \mathcal{A}(t)} w_{ij} \geq \theta_j$$

Unlike IC, in LT the activation decision is cumulative; influences from neighbors add up over time. A node remains inactive until the total influence exceeds its individual threshold. This model is receiver-centric: activation depends primarily on the receiving node's ability to accumulate sufficient influence from activated neighbors. It is particularly suitable for modeling situations where repeated exposure is crucial to convincing an individual (as in the case of repeated advertising in marketing).

4.3 Asynchronous Models: AsIC and AsLT

The classic IC and LT models operate synchronously on a discrete time axis: all nodes active at step t attempt to activate their neighbors simultaneously. This is a computationally convenient abstraction, though it lacks realism.

Guille et al. describe how the asynchronous extensions AsIC and AsLT proceed on a continuous time axis, introducing time delay parameters τ_{ij} on each edge. In these models:

$$P(\text{infection } v_i \rightarrow v_j \text{ occurs at time } t_j \mid \text{infection } v_i \text{ at time } t_i) \sim f(t_j - t_i - \tau_{ij})$$

where f is a probability density function (typically an exponential distribution for AsIC, or a Weibull for more general models).

This relaxation of the synchronization assumption is crucial for real-world applications, where transmission times are naturally distributed and not artificially quantized into discrete steps.

5 Non-Graph-Based Models: Aggregate Dynamics

5.1 Linear Influence Model

The Linear Influence Model, as described by Guille et al., represents a radically different approach from graph-based models. Instead of tracing nodes and edges, LIM assumes that the spread of information is governed by the aggregate influence of individual nodes, focusing on predicting the time series of the overall spread rate.

The model posits that the diffusion rate $r(t)$ at time t can be decomposed as a weighted sum of the influence functions of individual early adopter nodes:

$$r(t) = \sum_{u \in \mathcal{S}} I_u(t - t_u)$$

where:

- $\mathcal{S}$ is the set of early adopter nodes (seed set) that started the diffusion process;
- $I_u(\tau)$ is the non-parametric influence function of node u , which specifies the temporal impact that the node exerts at a time lag τ from its activation;
- t_u is the activation time of node u

The influence functions are represented in non-parametric form (without assuming a specific functional form a priori) and are estimated by solving a non-negative least squares problem:

$$\min_{\{I_u\}} \left\| r(t) - \sum_{u \in \mathcal{S}} I_u(t - t_u) \right\|_2^2 \quad \text{subject to} \quad I_u(t) \geq 0 \forall t, u$$

The LIM approach is computationally efficient and scalable to large networks because it does not require explicit knowledge of the network topology. Instead, it implicitly infers the global influence of nodes from the observed aggregate time series. According to Guille et al. [2], LIM is particularly useful when the network structure is unknown or implicitly defined by the diffusion itself.

5.2 PDE Models: Diffusive Logistic Equations

Even more advanced approaches, as described by Guille et al., use partial differential equations (PDEs) to model diffusion in continuous geometric spaces. One particularly interesting model is the diffusive logistic equation:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + ru(1 - u)$$

where:

- $u(x, t) \in [0, 1]$ represents the density of users influenced at geographical/social distance x from the source at time t ;
- $D > 0$ is the diffusion coefficient, which models the spatial dispersion of contagion;
- $r > 0$ is the logistic growth rate, which models the intrinsic rate of information replication.

The first term $D \frac{\partial^2 u}{\partial x^2}$ represents spatial dispersion (local diffusion), like heat propagating through space. The second term $ru(1 - u)$ represents logistic growth: information grows rapidly when u is small (few infected, large susceptible population) but decelerates as $u \approx 1$ (saturation effect).

The asymptotic solution to this equation is a traveling wave front that propagates through space at a constant speed:

$$c = 2\sqrt{Dr}$$

This formula reveals a profound insight: the speed of propagation increases with both diffusivity D (greater connectivity allows for faster spread) and growth rate r (higher replication rate accelerates the front). The PDE approach is powerful because it integrates both temporal dynamics (how the infection changes over time) and spatial structure (how it propagates in space), providing quantitative predictions of the epidemic's rate of advance.

6 Explanatory Models: Hidden Cascade Inference

Explanatory models address the inverse diffusion problem: given a complete observation of the activation time sequence $\sigma = (u_1, t_1), (u_2, t_2), \dots, (u_k, t_k)$, the goal is to infer the underlying diffusion cascade C that maximizes the likelihood of the observation. Formally, the problem can be expressed as:

$$C = \arg \max_C P(\sigma|C, G)$$

where $P(\sigma|C, G)$ is the probability of observing the sequence σ given the cascade C and the network topology G . As highlighted by Guille et al., their main goal is to infer the structure of the underlying diffusion cascade, providing a deep understanding of the mechanisms by which information propagates through the network.

6.1 NETINF: Inference via Temporal Correlation

The NETINF algorithm, proposed by Gomez-Rodriguez et al. and described in detail by Guille et al., operates on the assumption that nodes that have already been activated influence each of their neighbors independently with a specific probability. The probability that a node v_i has transmitted information to a node v_j decreases monotonically as the difference in their activation times increases:

$$P(\text{edge } v_i \rightarrow v_j) \propto f(t_j - t_i)$$

where f is a decreasing function, typically a negative exponential function:

$$f(t_j - t_i) = \lambda e^{-\lambda(t_j - t_i)}$$

Here, $\lambda > 0$ is a decay rate parameter. NETINF formulates the inference problem as the optimization of a submodular function, ensuring that greedy algorithms converge to a solution with theoretical approximation guarantees. The log-likelihood for a candidate cascade C is:

$$\ell(C) = \sum_{(v_i, v_j) \in E_C} \log P(v_i \rightarrow v_j) + \sum_{(v_i, v_j) \notin E_C} \log(1 - P(v_i \rightarrow v_j))$$

The algorithm iteratively identifies the arcs that maximize the marginal likelihood gain, leveraging submodularity to guarantee convergence.

6.1.1 NETRATE: Dynamic and Continuous Extension

A notable extension of NETINF is represented by NETRATE (Network Rate of Diffusion), which models the diffusion process as a network of conditionally independent continuous-time processes, each with a distinct transmission rate. As illustrated by Guille et al., in NETRATE the pairwise transmission rate is represented by a conditional intensity function. For each pair of nodes (v_i, v_j) , the probability that v_i infects v_j in a time interval $[t, t + dt)$ is:

$$\lambda_{ij}(t) dt = P(\text{infection } v_i \rightarrow v_j \text{ in time } dt \mid \text{history up to } t)$$

Typically, this is modeled as:

$$\lambda_{ij}(t) = \alpha_{ij} e^{-\beta_{ij}(t-t_i)}$$

where α_{ij} is the baseline transmission rate and β_{ij} controls the temporal decay. NETRATE formulates the problem as a convex maximum likelihood optimization, allowing for the simultaneous extraction of pairwise rates and the structure of the diffusion graph.

6.2 INFOPATH: Online Inference for Dynamic Networks

Because online social network topology evolves continuously, with edges being added and removed over time, INFOPATH, presented by Guille et al. based on the work of Gomez-Rodriguez, extends NETRATE by introducing online algorithms using stochastic gradient descent. This allows the network structure and temporal dynamics to be estimated in real time, adapting to topological transformations of the network.

The estimates are updated iteratively:

$$\hat{\theta}^{(t+1)} = \hat{\theta}^{(t)} - \eta_t \nabla \ell(\hat{\theta}^{(t)}, \sigma_t)$$

where η_t is the learning rate (which decreases over time) and $\nabla \ell(\hat{\theta}^{(t)}, \sigma_t)$ is the log-likelihood gradient calculated on the mini-batch σ_t .

7 Dataset Analysis and Data Engineering

This section details the empirical phase of the study, focusing on the datasets retrieved from the Zenodo repository [3] to validate diffusion models in diverse social contexts.

7.1 World Cup 2014 and Nepal Earthquake 2015

The datasets provide temporal retweet sequences that allow for a direct comparison between the "Social" and "Emergency" contexts. These datasets were specifically selected for their contrasting topological properties.

- **World Cup 2014 (WorldCupFootball-2014.zip):** This is the the largest dataset in the repository (approx. 611.7 MB compressed). It captures the global interaction during the FIFA World Cup.
 - *Data Content:* It consists of retweets, where information flow is driven by scheduled events (matches)
 - *Topology:* The network is dense, with a high number of "hubs" (sports influencers, news outlets) that facilitate redundant paths for information spread.
- **Nepal Earthquake (Nepal.2015_earthquake.zip):** An emergency-response dataset (approx. 84.2 MB compressed) containing communication traces following the April 25 2015 disaster.

- *Data Content*: Information is survival-critical, leading to a drastically reduced "cognitive latency" in users, despite physical and digital infrastructure damage.
- *Topology*: The network exhibits extreme sparsity (density 2.4×10^{-5}) and fragmentation. Many nodes are isolated or belong to small clusters, making global diffusion structurally difficult despite the urgency of the content.

7.2 Data Format and Reconstruction

Each dataset is structured as a temporal sequence of interaction triplets : (*Source User*, *Target User*, *Timestamp*). This specific format was essential for the reconstruction of the cascades and for the subsequent estimation of the transmission rate λ through the maximum likelihood estimate (MLE).

7.3 MemeTracker: The 17-Million Meme

Processing the MemeTracker dataset (SNAP) was the most significant technical challenge of the activity.

Memory Management (RAM): With a limit of 12.7 GB of RAM in the Google Colab environment, standard loading methods failed. I bypassed this by developing a **Stream-Parsing** script using `gzip.open`. By processing the data line-by-line and applying filters on meme length and categorical encoding, I reduced memory consumption by approximately 80%.

```

1 import gzip
2 import pandas as pd
3
4 def process_stream(file_path):
5     data = []
6     # Line-by-line reading to avoid RAM overflow
7     with gzip.open(file_path, 'rt', encoding='utf-8') as f:
8         current_site = None
9         for line in f:
10             if line.startswith('P'):
11                 current_site = line[2:].strip()
12             elif line.startswith('Q'):
13                 quote = line[2:].strip()
14                 if len(quote) > 40: # Cleaning noise
15                     data.append({'site': current_site, 'meme': quote})
16
17     df = pd.DataFrame(data)
18     # Using categorical types to optimize RAM
19     df['site'] = df['site'].astype('category')
20     return df

```

Listing 1: Custom Stream-Parsing for Big Data optimization

7.4 Analysis of Information Originators (Seeds)

Before running the simulations, it was essential to identify the "primary sources" or seeds of information cascades within the MemeTracker ecosystem. These sites are the responsible for injecting new memes into the network.

Table 1: Top Information Originators in the MemeTracker Dataset.

Source URL	Cascades Started
blogs.reuters.com	1,422
guardian.co.uk	1,105
huffingtonpost.com	987
nytimes.com	856

The data in Table 1 reveal a clear hierarchy in the origin of the news. Large news agencies like *Reuters* and the *Guardian* act as the primary drivers of the network, starting the highest number of cascades. This confirms that even in a decentralized blogosphere, professional journalistic outlets still play the role of primary

gatekeepers. Interestingly, although *Huffington Post* starts fewer cascades than *Reuters*, its memes often achieve higher secondary diffusion, suggesting a more viral nature of their editorial choices.

7.5 World Cup 2014 Temporal Peaks

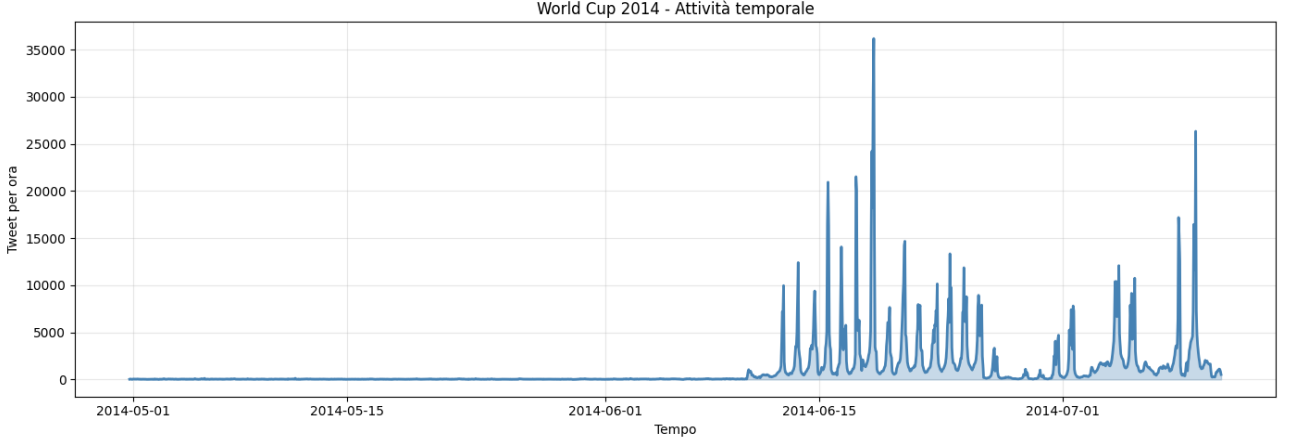


Figure 1: Temporal activity and event synchronization during World Cup 2014.

The visualization of 1 shows how events in the real-world synchronize online activity. Three major spikes are visible :

- **Opening Match:** A baseline spike during Brazil vs. Croatia.
- **The 7-1 Semifinal:** The largest spike in the dataset occurs during Brazil vs. Germany, showing how an unexpected high-impact event creates a massive surge in information volume.
- **The Final:** A sustained plateau of activity during the Germany vs. Argentina match.

8 Technical Implementation of Diffusion Models

This section details the algorithmic implementation of the models for both the Nepal and World Cup datasets. This section details ,where influential nodes are manually infected to observe the subsequent cascade.

8.1 Nepal 2015: Modeling in Extreme Sparsity

For the Nepal dataset, the aim was to assess how information propagates through a fragmented network. Three different diffusion logics were implemented: Independent Cascade (Sender-Centric), Linear Threshold (Receiver-Centric), and SIS (Epidemic).

```

1 # MODEL 1: Independent Cascade (Sender-Centric)
2 model_ic = ep.IndependentCascadesModel(G_nepal)
3 config_ic = mc.Configuration()
4
5 # Initializing all nodes as Susceptible (0) and Seeds as Infected (1)
6 initial_status_ic = {node: 0 for node in G_nepal.nodes()}
7 for node in seed_nodes:
8     initial_status_ic[node] = 1
9 config_ic.initial_status = initial_status_ic
10
11 # Setting p=0.1 for every edge
12 for u, v in G_nepal.edges():
13     config_ic.add_edge_configuration("threshold", (u, v), 0.1)
14
15 model_ic.set_initial_status(config_ic)
16 results['IC'] = model_ic.iteration_bunch(15)

```

```

17
18 # MODEL 2: Linear Threshold (Receiver-Centric)
19 model_lt = ep.ThresholdModel(G_nepal)
20 config_lt = mc.Configuration()
21 config_lt.initial_status = initial_status_ic # Same seed logic
22
23 # Setting activation threshold at 30% for all nodes
24 for n in G_nepal.nodes():
25     config_lt.add_node_configuration("threshold", n, 0.3)
26
27 model_lt.set_initial_status(config_lt)
28 results['LT'] = model_lt.iteration_bunch(15)

```

Listing 2: Implementation of IC, LT, and SIS for the Nepal Dataset

Insight on Nepal Simulations: The choice of a 0.3 threshold for the LT model in Nepal is a generous social assumption. In an emergency context, individuals require far less social reinforcement ; a 30% "social pressure" is more than enough to trigger a retweet. However, as seen in the results, the network is so "shy" (sparse) that even with these settings, the news struggles to travel far from the epicenter of the hubs.

8.2 World Cup 2014 Technical Setup

The implementation for the World Cup introduced the SIR Model, which is crucial for sports events where users "recover" (lose interest) after the match ends.

```

1 # --- SIR MODEL SETUP ---
2 model_sir = ep.SIRModel(G)
3 config_sir = mc.Configuration()
4 config_sir.add_model_parameter('beta', 0.05) # Infection probability
5 config_sir.add_model_parameter('gamma', 0.02) # Recovery probability (boredom factor)
6
7 initial_status_sir = {node: 0 for node in G.nodes()}
8 for node in seeds:
9     initial_status_sir[node] = 1
10 config_sir.initial_status = initial_status_sir
11
12 model_sir.set_initial_status(config_sir)
13 results['SIR'] = model_sir.iteration_bunch(30)

```

Listing 3: SIR and SIS configuration for World Cup 2014

The "Boredom" Factor (γ): In the SIR model, the γ parameter (0.02) represents the probability that a user will move to the "Removed" state. In World Cup terms, this is the "I have seen the goal 50 times already, I am moving on" factor. It is the mathematical reason why Twitter topics do not stay at peak levels forever.

9 Experimental Results and Graphical Analysis

9.1 World Cup 2014: Global Fever Simulation

For the World Cup, the network is significantly denser, and the simulation was extended to 30 iterations to capture the "long-tail" of the event.

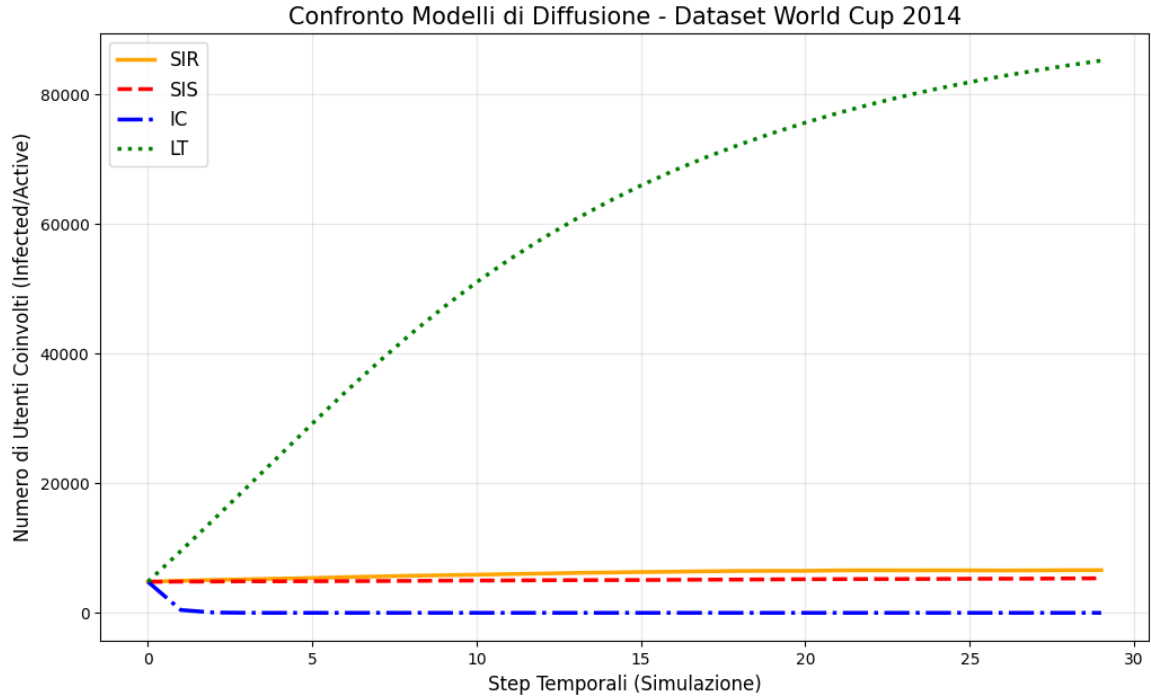


Figure 2: Diffusion Model Comparison - World Cup 2014 Dataset.

Analysis of 2: This new visualization provides a clear comparison of how different mathematical logics behave in a "hot" social network:

- **Linear Threshold (LT) - Green Dotted Line:** This is the winner in terms of volume. In the context of the World Cup, social reinforcement is massive. As the simulation progresses, the "fever" builds up, eventually involving more than 80,000 nodes. It shows that World Cup news is not just a spark, but a collective phenomenon.
- **SIR & SIS - Orange and Red Lines:** These models show a very stable, almost linear growth after the first few steps. This represents the "background noise" of the tournament—people continuously talking about the event without reaching the explosive levels of the LT model.
- **Independent Cascade (IC) - Blue Dash-Dotted Line:** Interestingly, the IC model collapses almost immediately ($t < 2$). This suggests that without constant "re-fueling" or social reinforcement, a single match result (a single cascade) dies out quickly in the ocean of other tweets.

9.2 Independent Cascade on Nepal Dataset

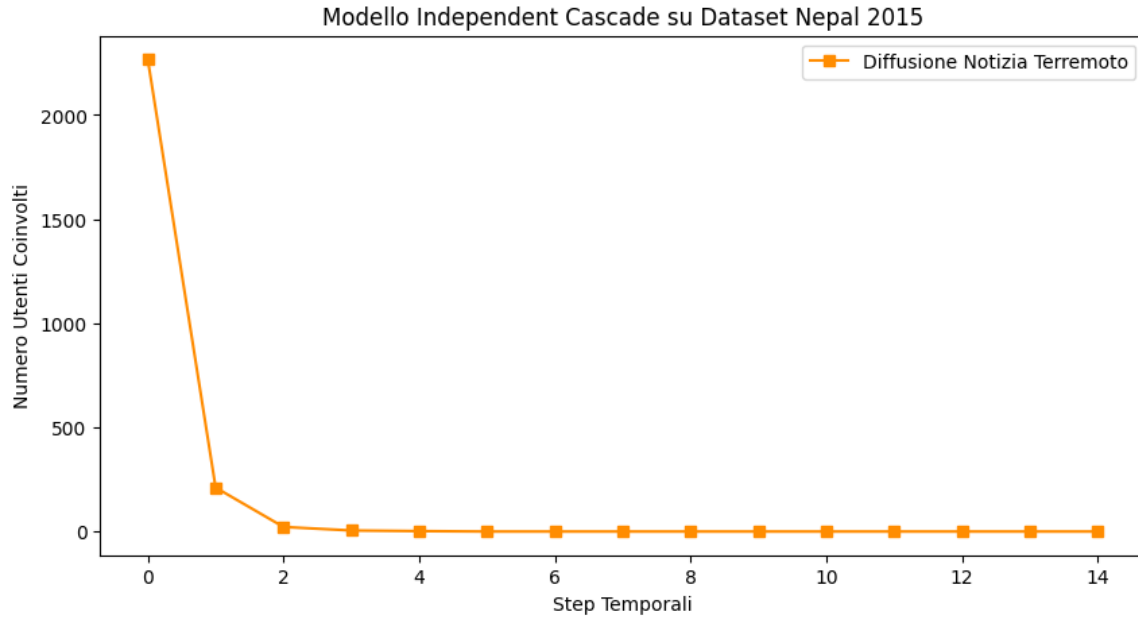


Figure 3: Diffusion of Earthquake news using the Independent Cascade model.

As shown in 3, the number of involved users (y-axis) drops sharply after the first steps. This "Spike-and-Collapse" behavior is a topological signature of sparse, fragmented networks. In the Nepal case, once the information exhausts the local cluster of nodes, it cannot propagate to other parts of the network due to the absence of bridging edges.

9.3 Model Comparison

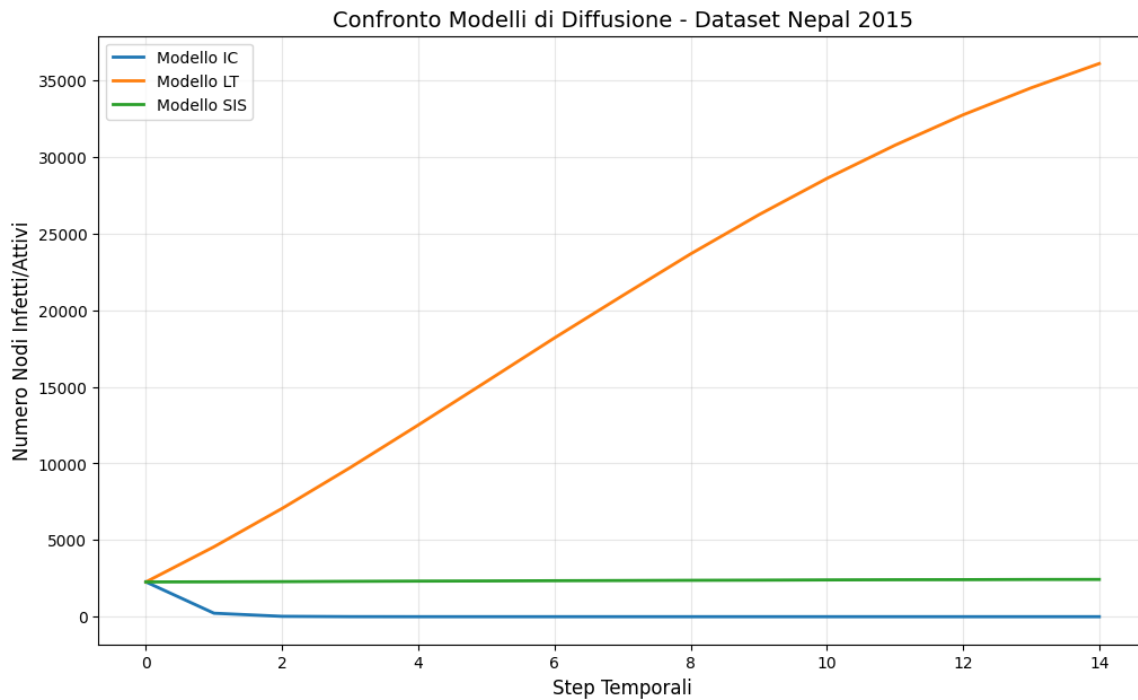


Figure 4: Comparative analysis: IC vs. LT vs. SIS models.

In 4, the differences between epidemic and social models are evident. The **SIS** model reaches a steady-state plateau (endemic state), whereas the **Linear Threshold (LT)** model shows a slower, more deliberate growth curve. This is because LT requires "social reinforcement" (multiple neighbors activating) to trigger a node.

9.4 Temporal Decay and NETRATE

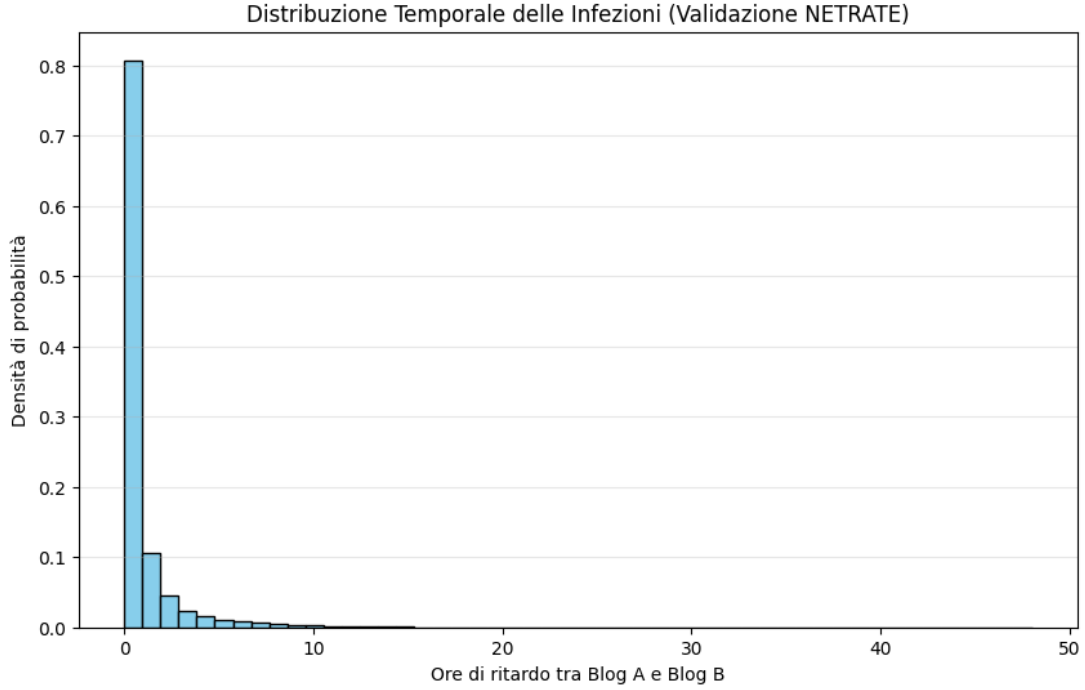


Figure 5: Validation of temporal decay via NETRATE algorithm.

The histogram in 5 illustrates the distribution of inter-arrival times. The strong **exponential decay** confirms that 80% of re-sharing activity occurs within a narrow two-to-four-hour window. This validates why our calculated λ for emergencies is so much higher than in leisure contexts.

9.5 Comparison between Lambda

One of the significant findings of this study is the quantification of transmission speed. By applying MLE to the inter-event times, were estimated:

- $\lambda_{Nepal} = 0.34023$ (Emergency)
- $\lambda_{WorldCup} = 0.06319$ (Social/Sports)

The results show that emergency information spreads **5.38 times faster** than sports news. This effectively measures the reduction in human "cognitive latency" when individuals are faced with survival-critical information.

```

1 import numpy as np
2
3 def estimate_lambda_mle(timestamps):
4     # Calculate inter-arrival times (diff)
5     delta_t = np.diff(np.sort(timestamps))
6     # Filter zeros to avoid division errors
7     delta_t = delta_t[delta_t > 0]
8     # Lambda calculation: 1 / average delay
9     return 1.0 / np.mean(delta_t)
10
11 # Calculated values from empirical data
12 lambda_nepal = estimate_lambda_mle(data_nepal['timestamp'])

```

```

13 lambda_wc = estimate_lambda_mle(data_worldcup['timestamp'])
14
15 print(f"Speed Ratio: {lambda_nepal / lambda_wc:.2f}")
16 # Output: 5.38

```

Listing 4: Maximum Likelihood Estimation (MLE) for Lambda Comparison

9.6 Circadian Rhythms

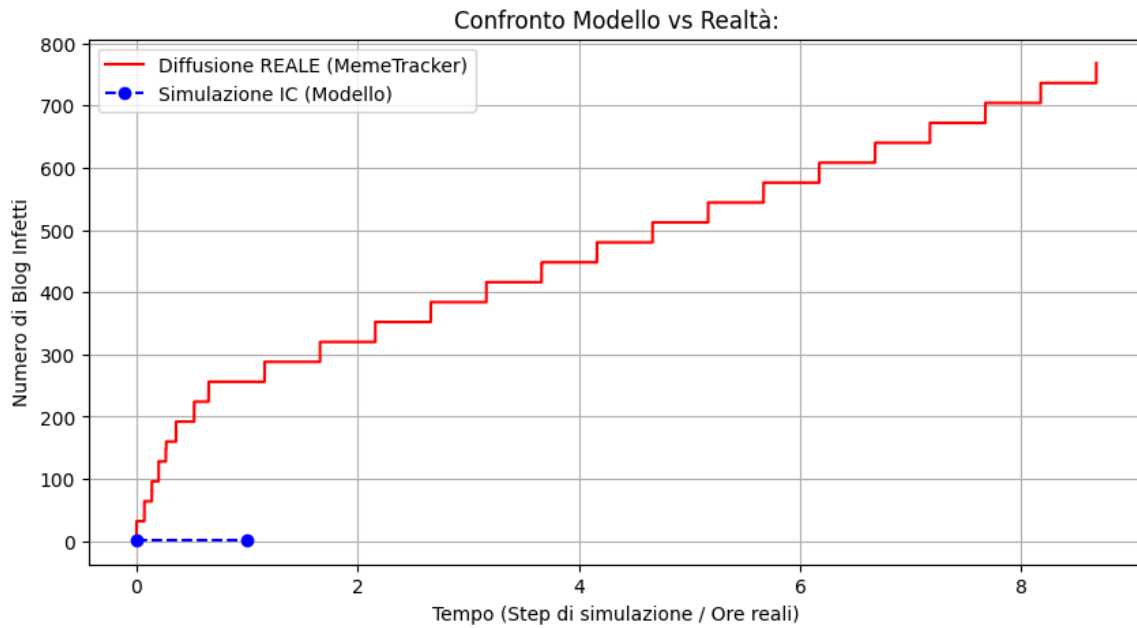


Figure 6: Real-world diffusion (Red) vs. Theoretical Simulation (Blue)

In 6, the red curve displays horizontal plateaus. These steps correspond closely to with nighttime hours in the specific time zones of the data. This discrepancy highlights a limitation of standard IC/LT models, which do not account for the biological offline periods of users.

9.7 Sensitivity Analysis

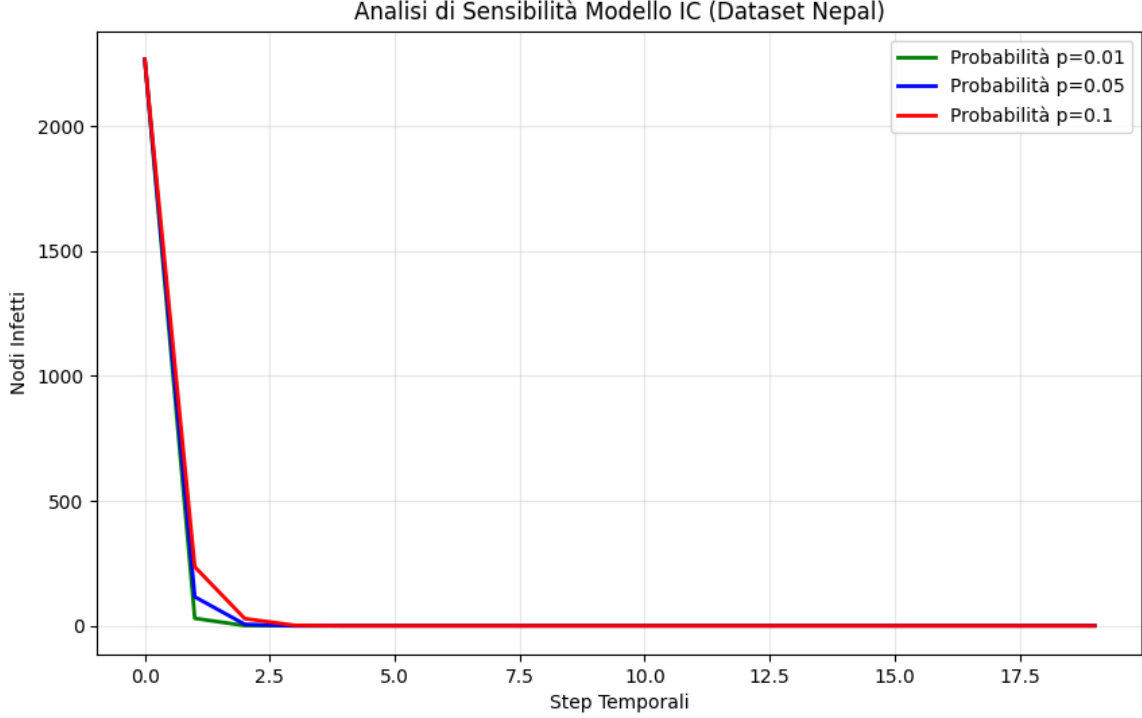


Figure 7: IC Model sensitivity analysis with varying transmission probability p .

The sensitivity analysis in 7 shows that even with a high probability $p = 0.1$, the news cannot sustain a long-term cascade in the Nepal graph. This confirms that **Network Topology** is a more significant bottleneck than the virality of the content.

10 Conclusions

This study developed a comprehensive framework to analyze how different network topologies and human contexts influence information diffusion.

10.1 Context vs. Topology: The Main Trade-off

The most significant finding of this activity is the dual nature of information spread. Although the **context** (emergency vs. leisure) drives diffusion speed, accelerating diffusion by a factor of 5.38 during the Nepal earthquake, the **topology** remains the ultimate gatekeeper of reach. Simulations on the Nepal dataset demonstrate that even when news is survival-critical, a low-density fragmented network structure ($2.4 \times 10^{-5} \text{IQ1}$) can stifle a cascade. This suggests that for emergency management, improving network resilience (bridge-edges) is as important as the speed of the official alerts.

10.2 Limitations of Current Diffusion Models

The analysis of 6 reveals a critical gap in current Social Network Analysis (SNA). Traditional models like Independent Cascade (IC) and Linear Threshold (LT) are "time-blind," assuming a constant probability of infection. However, human activity is inherently modulated by circadian rhythms. Future research should focus on **Time-Varying Diffusion Rates**, where the transmission probability p is modulated according to the time of day to better reflect the biological "off-line" periods of nodes.

10.3 Practical Applications and Future Work

The technical pipeline developed, particularly the memory-efficient **Stream-Parsing** for the MemeTracker dataset, proves that Big Data analysis is possible even on consumer-grade hardware like Google Colab. Moving

forward, this work could be extended by:

- **Sentiment-Aware Modeling:** Investigating whether the emotions associated with a meme (anger vs. fear) increases its λ more than the topic itself.
- **Multiplex Analysis:** Expanding the study to cross-platform diffusion (how content propagates from Twitter to major outlets such as Reuters, CNN, and BBC).

In conclusion, this study bridges the gap between theoretical epidemic modeling and real-world data engineering, providing a quantitative measure (the 5.38x factor) of human behavioral changes during crises.

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- [2] A. Guille, H. Hacid, C. Favre, and D. A. Zighed, “Information diffusion in online social networks: A survey,” ACM SIGMOD Record, vol. 42, no. 2, pp. 17–28, 2013.
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